

EV Charging Station Monitoring Platform Using AWS

1. Project Overview

The EV Charging Station Monitoring Platform is a cloud-based solution built on AWS. It monitors charging station status, power consumption, and temperature in real time. The platform uses AWS IoT Core, Lambda, DynamoDB, API Gateway, SNS, S3, and CloudWatch.

2. Architecture and AWS Services

AWS IoT Core receives charger data through MQTT topics. Lambda processes incoming data. DynamoDB stores station information. SNS sends alerts. API Gateway exposes data to applications. S3 hosts the dashboard and CloudWatch provides monitoring and metrics.

3. Implementation Steps

Create a DynamoDB table, configure AWS IoT Core Things and certificates, create MQTT topics, build Lambda functions, configure IoT Rules, set up SNS alerts, create API Gateway endpoints, and host the frontend dashboard using Amazon S3.

4. Testing and Monitoring

Publish test MQTT messages and verify records in DynamoDB. Confirm Lambda execution, CloudWatch logs, API responses, and SNS email alerts. Monitor charger performance using CloudWatch dashboards and metrics.

5. Results and Conclusion

The platform successfully provides real-time monitoring of EV charging stations. It improves operational visibility, supports proactive maintenance through alerts, and demonstrates serverless AWS architecture suitable for scalable IoT applications.